INCH-POUND

MIL-M- 87996 (84)

15 Feb 1990

MILITARY SPECIFICATION

MIXER, CONCRETE, MOBILE, 5 CUBIC YARD TRUCK MOUNTED

This specification is approved for use within WR-ALC Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 This specification covers a truck-mounted, 5 cubic yard, mobile concrete mixer.

2. APPLICABLE DOCUMENTS.

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadmium (Electodeposited)
KKK-T-2109	Trucks and Truck Tractors: Commercial, Diesel Engine Driven, 34,500 to 66,000 Pounds GVW, 6x4

Military

MIL-P-514	Plates, Identification, Instructions and Marking, Blank
MIL-T-704	Treatment and Painting of Material

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMVRG, Robins AFB, GA 31098-5609 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 3895

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MIL-B-7883	Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-P-23377	Primer Coating, Epoxy-Polyamide, Chemical and Solvent Resistant
MIL-P-26915	Primer Coating, Zinc Dust Pigmented, for Steel Surfaces
MIL-I-46058	Insulating Compound, Electrical (For Coating Printed Circuit Assemblies)
MIL-C-46168	Coating, Aliphatic Polyurethane, Chemical Agent Resistant
MIL-S-81733	Sealing and Coating Compound, Corrosion Inhibitive
MIL-T-83133	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8
MIL-H-83282	Hydraulic Fluid, Fire Resistant, Synthetic, Hydrocarbon Base, Aircraft
MIL-C-83286	Coating, Urethane Aliphatic Isocyanate, for Aerospace Applications

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-12	Abbreviations for Use on Drawings and on Specifications, Standards, and Technical Documents
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-209	Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment
MIL-STD-248	Welding and Brazing Procedure and Performance Qualification
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-721	Definitions of Terms for Reliability and Maintainability
MIL-STD-808	Finishes, Materials, and Processes for Corrosion Prevention and Control in Support Equipment
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-970	Standards and Specifications, Order of Preference for the Selection of
MIL-STD-1223	Nontactical Wheeled Vehicle Treatment, Painting, Identification Marking, and Data Plate Standards
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment, and Facilities
MIL-STD-1474	Noise Limits for Army Material
MIL-STD-1568	Materials and Processes for Corrosion Prevention and Control in Aerospace Weapon Systems
MIL-STD-1587	Materials and Process Requirements for Air Force Weapon Systems
MIL-STD-1791	Designing for Internal Aerial Delivery in Fixed Wing Aircraft

Handbooks

MIL-HNBK-781	Reliability Design Qualification and Production Acceptance Tests: Exponential Distribution

(Unless otherwise indicated, copies of the federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

US Air Force (USAF) Drawings

8838302	Fastener Quality Assurance Requirements for WR-ALC/MMV Vehicle and Support Equipment

(Drawing is available from WR-ALC/MMDEC, Robins AFB, Ga. 31098-5609)

NATO STANDARD DIESEL AND SPARK IGNITION ENGINES LABORATORY TEST
AEP-5 test procedures

(Application for copies should be addressed to the US Army Tank and Automotive Command, ATTN: AMSTA-RGRD, Warren, MI 48397-5000.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

American Society of Mechanical Engineers (ASME)

Welding Qualifications

Section 9-86

(Application for copies should be addressed to American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York NY 10017.)

American Welding Society (AWS)

Welding Qualifications

B2.1-84

(Application for copies should be addressed to American Welding Society, 2501 NW Seventh Street, Miami FL 33125.)

Society of Automotive Engineers (SAE)

J336a	Sound Level for Truck Cab Interior
J360	Motor Vehicle Grade Parking Performance Test Code
J366b	Exterior Sound Level for Heavy Trucks and Buses
J534	Lubrication Fittings
J695	Turning Ability and Off Tracking – Motor Vehicles
J786a	Brake System Road Test Code – Truck, Bus, and Combination of Vehicles
J994	Alarm – Backup – Electric – Performance, Test and Application
J1077	Measurement of Exterior Sound Level of Trucks with Auxiliary Equipment

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(Application for copies should be addressed to Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale PA 15096.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS.

3.1.1 Selection of specifications and standards. Specifications and standard, necessary commodities, and services not specified herein shall be selected in accordance with MIL-STD-970.

3.1.2 Description. This specification covers a truck-mounted, mobile concrete mixing plant capable of transporting unproportioned and unmixed materials for producing 5 cubic yards of mixed concrete. The mixer shall be mounted on a commercial truck chassis as specified herein (see 3.3). The mixer shall be complete with all components furnished as standard equipment that are customarily offered with a mixer of this type whether stipulated herein or not, together with such attachments as may be necessary to enable the mixer to function reliably and efficiently in a sustained operation. Established component ratings shall not be increased to meet the requirements of this specification.

3.1.3 Materials. The selection of materials and processes shall be in accordance with MIL-STD-1568 and MIL-STD-1587. Materials not covered by this specification or referenced document shall be suitable for the intended environment and application. The contractor shall not use nonstandard or proprietary materials and processes without prior government approval. The components, pieces and parts incorporated in the mixer may be newly fabricated from recovered materials to the maximum extent practical, provided the mixers produced meet all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the mixer. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinquished from virgin raw materials.

3.1.3.1 Protective treatment. Materials that are subject to deterioration shall be protected in such a manner that shall in no way prevent compliance with the performance requirements of this specification. Protective coating that will chip, crack, or scale with age or extremes of climatic conditions shall not be used. Surface preparation, finishes and coatings shall be in accordance with MIL-STD-808. The contractor shall prepare and submit for government review, a corrosion prevention and control plan (CPCP). The CPCP shall include, as a minimum, the contractor's methodology for incorporating corrosion prevention and control at the design level and a surface preparation and finish specification for all materials in accordance with MIL-STD-1568, paragraphs 5.1.1 and 5.1.2. Submittal of above documents does not constitute release from any specification requirements.

3.1.3.2 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.1.3.3 Primers. MIL-P-23377 and MIL-P-26915 epoxy primers shall be used on aluminum and ferrous alloys respectively.

3.1.3.4 Topcoat. MIL-C-83286 polyurethane shall be used as the topcoat finish.

3.1.3.5 Drainage. Complete drainage shall be provided for both internal and exterior pockets, areas, parts and components.

3.1.3.6 Magnesium. Magnesium and magnesium alloys shall be subject to approval by the Government.

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3.1.3.7 Prohibited materials. The use of wood and/or wood products is prohibited. Similarly, vinyl, polyvinyl chloride, polyester, and corrosive type RTV (yields acetic acid upon curing) are prohibited for use in all forms, interior and exterior.

3.1.3.8 Fasteners. Ferrous alloy fasteners shall be cadmium plated per QQ-P-416 Type II, to a minimum thickness of 0.0003 inches unless threaded tolerances dictate a different thickness, and aluminum alloy fasteners shall be anodized per MIL-A-8625. Fastener installation shall be in accordance with MIL-STD-1568.

3.1.3.9 Sealants for permanent fasteners. All joints, rivets and mating/faying surfaces of exterior and interior components shall be sealed using MIL-S-81733.

3.1.3.10 Conformal coating. Printed circuit/printed wire boards shall be conformal coated with materials in accordance with MIL-I-46058.

3.1.3.11 Commercial off-the-shelf equipment. All commercial equipment and parts shall be assessed by the contractor as to their suitability to function in the intended environment and comply with the system's life expectancy requirements.

3.1.3.12 Galvanized steel. Many producers of zinc-coated steel apply inhibitors to protect their products from moisture during shipment and storage. Some of these inhibitors interfere with pretreatments resulting in poor priming surfaces. Contractors shall consult with their supplier of galvanized steel for recommendations on the pretreatment to be used on their respective product.

3.1.4 Design and construction. The vehicle shall be constructed so that parts do not work loose in service and shall withstand the loads, shocks, vibrations, and other conditions incident to operation, shipping, and storage with minimum loss of time for maintenance, repair, and servicing. The design of the vehicle shall ensure safe, efficient, and economical operation. All assemblies, controls, and installed equipment shall be located so that there is no adverse interference with each other, or with the operation, and shall be readily accessible for maintenance, operation, and replacement.

a. Finishes. Protective coating application shall be considered during the design stage. Sharp, angular corners are undesirable due to coating failures on outside corners (too thin) and the tendency of coatings to bridge inside corners. Smooth, round corners allow uniform coating thickness and avoids creation of possible sump areas. All mechanically joined components shall be finished before assembly. Welded structures shall have finishes applied after welding. When welded and bolted assemblies are used, manufacturing steps shall be arranged to minimize disruption of the protective coatings. Supplementary corrosion prevention treatments - rustproofing, sealants, greases, etc., shall be applied after finish application. All metallic surfaces shall be treated for corrosion prevention.

b. Joints and faying surfaces. Joint designs shall minimize crevices formed by overlapping surfaces and shall point away from the forward direction of vehicle travel or point down. Butt welds are preferable to lap welds. Continuous welds are preferable to tack welds. The joints shall be designed so there are no gaps on joint laps. This can be accomplished by continuous welds, solder or sealant bead lines. All faying surfaces shall be primed and completely coated with sealant prior to assembly.

c. Core panel structures. Pretreat and prime all internal surfaces prior to assembly. Pretreat and prime interior panel skin and faying surfaces prior to assembly. Coat entire faying surface area with sealant during assembly and apply sealant bead after assembly to the lap joint. The panels shall then be foamed in place using non-reverting polyurethane.

d. Drain holes. Structures shall be designed to avoid pockets where debris or moisture might accumulate. Inverted u-channel structural members do not accumulate moisture or debris. Closed structural configurations should have sloping surfaces towards drain holes which are adequately sized and placed in the lowest sump areas. All interior surfaces shall be treated in accordance with MIL-STD-808, with priming being the minimum finish requirement. The finish shall be reapplied/applied through drain holes after welding operations.

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e. Alternate materials. Use of composites, polymers, etc., type materials should be investigated. While not subject to corrosion, consideration of material interaction, environmental stability and field repair should be investigated before recommending alternate material.

3.1.4.1 Reliability. The vehicle, including all components, shall have an observed Mean-Time-Between-Failure (MTBF) of not less than 100 hours. The vehicle shall also have an operational availability of not less than 0.99. A failure is defined as the event or inoperable state in which any item or part of an item does not, or would not, perform as previously specified. Operational availability (Ao) is defined as the percent of time that the system is capable of satisfactorily performing in the operational environment.

Mathematically Ao is expressed as:

Ao = MTBF/(MTBF + MRT),
where MRT is Mean Repair Time.

Mean repair time (MRT) is the elapsed time to accomplish all repairs divided by the number of repairs performed. Mathematically, MRT is expressed as:

MRT = T/N,
where T is Elapsed Time
N is Number of Repairs.

The elapsed time includes all maintenance actions required to correct the malfunction, including time to prepare, remove, replace, repair, adjust, and check. MRT does not include administrative or supply delays. Note that time to troubleshoot is not included; thus, this is not the standard definition for MRT.

Definitions of reliability terms not included herein shall be as found in MIL-STD-721 and MIL-HDBK-781. In the event of a conflict between these documents, MIL-HDBK-781. In the event of a conflict between these documents, MIL-HDBK-781 shall take precedence. The vehicle shall be designed for a life expectancy of at least 10 years.

3.1.4.2 Maintainability. The vehicle shall be designed and constructed to provide the following:

a. A minimum number of parts consistent with the performance required herein.
b. A minimum amount of time for assembling, disassembling, locating trouble sources, and maintaining the vehicle. Where practical, components shall be located for rapid inspection and recognition of potential failure.
c. Ease of adjusting, servicing, and replace parts and components.
d. Use of readily availiable standard tools and equipment for maintenance.
e. Maintenance with a minimum number of tools.
f. Maximum repair time, excluding time to troubleshoot, of no greater than eight hours.
g. Maximum time to perform all daily scheduled maintenance and servicing tasks of no greater than 15 minutes.

3.1.4.3 Maintenance tasks. The vehicle shall be designed so that one mechanic, using common hand tools, can perform the following maintenance operations within the time specified:

a. Remove, replace, and adjust all engine driven belts - 45 minutes.
b. Remove and replace alternator - 30 minutes.
c. Remove and replace all filters, screens, and strainers in all hydraulic systems - one hour.
d. Remove and replace engine coolant system hoses - 30 minutes.
e. Drain engine lubricating oil, remove, and replace oil filter elements and refill crankcase - one hour.
f. Remove and replace fuel filter element(s) - 30 minutes.

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g. Disconnect battery cables, remove and replace batteries and reconnect battery cables - 30 minutes.
h. Drain torque converter oil and transmission oil, remove and replace all filter elements and strainers, and refill converter and transmission - one hour.
i. Remove and replace starter - one hour.
j. Remove or install the engine/transmission package, two mechanics - four hours.

3.1.4.4 Fastening devices. All screws, bolts, nuts, pins, and other fastening devices shall be properly designed, manufactured, and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers, or nylon patches shall not be used for this purpose, except for attachment of trim items or as provided in commercial components. Tapped aluminum threads shall have a minimum thread engagement of at least two times the nominal fastener diameter. All fasteners of Grade 5 or higher must meet the quality assurance requirements of US Air Force Drawing 8838302.

3.1.4.5 Servicing provisions. Drains, grease fittings, lines, and check points for all components shall be so located that they are readily accessible and do not require special tools for proper servicing.

3.1.4.6 Lubrication. All parts normally requiring grease for lubrication shall be provided with lubrication fittings in accordance with SAE J534. Where the use of high lubrication pressures may damage grease seals, a pressure relief device shall be provided. Extended fittings shall be provided to lubricate parts of assemblies which are not readily accessible for direct lubrication or which are likely to be overlooked because of inaccessibility.

3.1.4.7 Hydraulic fluid. Hydraulic fluid shall be in accordance with MIL-H-83282.

3.1.4.8 Component protection. All space in which work is performed during operation, service, and maintenance of the vehicle shall be free of hazardous protrusions, sharp edges, or other features which may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard.

3.1.4.9 Foreign object damage. All loose metal parts, such as pins and valve caps, shall be securely attached so they cannot become separated from the vehicle.

3.1.5 Environmental conditions. The vehicle shall function and travel as specified herein under worldwide climatic conditions from -25°F to +125°F and from sea level to 5000 foot altitude. The mixing plant shall function between +32°F to +125°F.

3.1.6 Air transportability. The vehicle shall be air transportable without shoring in accordance with MIL-STD-1791 in Types C-130, C-141, and C-5 Aircraft and shall be air transportable in a Type C-17 Aircraft.

3.1.6.1 Equipment removal. Removal of equipment shall be permitted, if necessary, to provide the required height and width for air transport. Removal and reinstallation of equipment shall be limited to removal and attachment of standard fasteners and fittings with common hand tools. The weight of individual parts or assemblies which must be handled or positioned shall not exceed the limits specified in MIL-STD-1472 for a one person lift. All equipment removed shall be stored on the vehicle; all necessary caps and plugs shall be provided to permit driving and storage of the vehicle while in the air transport configuration. Removal or reinstallation of the equipment shall not take longer than 15 minutes for two mechanics using common hand tools.

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3.1.6.2 Tiedowns. The vehicle shall be equipped with sufficient tiedown points so located that it can be restrained in all directions during air transport. The attachment points shall be designed for use with attachment in the aircraft as described in MIL-STD-1791 and MIL-STD-209. Sufficient clearance shall be maintained around all attachment points to allow for restraining chains without interference from any other structure. The tiedown points shall have a minimum clear opening of three inches to allow attachment of a chain with a grabhook.

3.1.6.3 Slinging provisions. Slinging provisions shall be in accordance with MIL-STD-1791 and MIL-STD-209.

3.2 Mixer construction.

3.2.1 Performance. The mixer shall be capable of continuously proportioning and mixing quality concrete at a rate of 1 cubic yard per 2.5 to 4 minutes depending on mix design.

3.2.2 Capacity. The mixer shall meet the following physical characteristics:

Minimum carrying capacity	– cement	– 32 cubic feet
	– sand	– 75 cubic feet
	– aggregate	– 90 cubic feet
	– water	– 220 gallons

Each material shall be carried in a different compartment.

3.2.3 Bins and tanks. The bins and tanks shall have the capacities specified in 3.2.2. The sand and aggregate bins shall have open tops and shall be capable of being charged by front end loaders or overhead bins. The sand and cement bins shall each be equipped with a pneumatic vibrator to assure continuous flow of materials. The cement bins shall be of the close-top type. A tarpaulin of sufficient size to cover the sand and aggregate bins shall be provided. Tiedown points of the bin and the tarpaulin shall be provided. The sand and aggregate bins shall be covered with screens to keep oversize rocks from entering the bins and mixing area.

3.2.4 Quick loading cement hopper. A hopper shall be provided to fit over the standard opening in the top of the cement bin. The hopper shall be provided with a vibrating screen and hinged cover. The cover shall cover the entire hopper opening. The hinged cover shall be provided with a circular opening to accept and elephant trunk from an overhead bin. The opening shall be provided with a hand operable, removable cover.

3.2.5 Water system. Water shall be supplied by a minimum of 220 US gallon mounted tank. A water fill cap secured to the tank by a chain shall be provided. The inside diameter of the fill port shall not be less than 2 inches and the fill port shall be labeled "Water". A water level gauge of the sight tube type, graduated in US gallons, shall be provided and this gauge shall be mounted or protected in a manner which prevents breakage. A water pump shall be provided with the water system. A washout hose shall be provided to clean the mixing auger and other critical areas. A water control valve calibrated in US gallons shall be provided to control the amount of water being added to the mix.

3.2.6 Mixing system. Mixing of the concrete shall be by an auger in a reinforced rubber chute. The auger shall be equipped with replaceable wear plates. The mixing auger shall be powered. A system shall be provided to raise and lower the mixing chute. The mixing chute shall be equipped with an antisplash cover to facilitate clean-out and provide clean operation. Two delivery chutes shall be provided with supports and hand operable fasteners to attach to the mixing chutes.

3.2.7 Mix proportioning system. The continuous proportioning of materials shall be controlled by volume. Adjustable volume control gates shall be furnished on the sand and aggregate bins for proportioning the mix to meet desired design strengths. The cement shall be proportioned by a rotary feeder encased in a drop-

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through housing. The sand and aggregate shall be conveyed to the mixing chamber from the respective bins by a common conveyer belt. The conveyer belt shall be moved by two paralleled chains adequately supported by removable wear bars. A system to continuously oil the chains during mixing shall be provided. The chain oiling system shall be in a readily accessible location.

3.2.8 Mixer drive trains. Power for the mixer shall be provided from the truck engine or transmission through a power take-off. Power take-off speed shall be proportional to engine speed. The power take-off shall permit engagement without clashing with the engine running and transmission in neutral. The power from the power take-off shall be transmitted to power the hydraulic pump and water pump. Power take-off and controls are provided by the mobile mixer manufacturer.

3.2.9 Quick-load cement bin. A portable bin shall be provided into which bags of cement may be emptied. The bin shall be so designed that it may be lifted by a front end loader and the cement in it may be dumped into the cement bin on the mixer. The quick-load cement bin shall have a capacity of at least 32 cubic feet.

3.2.10 Admixture system. The manufacturer shall supply his commercial liquid and dry cement admixture systems.

3.2.11 Air system. The system provided to supply air pressure for the pneumatic vibrators shall be taken from the air brake system of the truck chassis. Assure that low pressure release of truck air brakes will not occur.

3.3 Truck chassis. The mixer shall be mounted on a commercial truck chassis as specified in KKK-T-2109. In the event of conflict between the text of this specification and KKK-T-2109, this specification shall take precedence. In addition to the engine requirements of KKK-T-2109, the vehicle engine shall operate on JP-8 fuel conforming to MIL-T-83133 without detrimental effect to its components. The engine shall be certified as having successfully completed NATO AEP-5 test procedures (400 hour dynamometer test) to assure JP-8 compatibility.

3.4 Finish and protective coatings.

3.4.1 Cleaning, painting, plating, anodic films, and chemical treatments. Cleaning, painting, plating, anodic films, and chemical treatments shall be in accordance with MIL-STD-808 and/or MIL-T-704, except that commercial components may be prepared in accordance with the manufacturer's standard practice, provided is it compatible with the exterior finish paint.

3.4.2 Finish. The exterior paint finish shall show no evidence of uneven application, curtains, runs, sags, orange peel, lack of adhesion, or other defects and shall be capable of passing the test described in paragraph 4.8.10 of MIL-C-83286.

3.4.2.1 Forest Green. Unless otherwise specified (see 6.2), the outer surface of the vehicle and the interior surfaces of compartments (except in the cab) shall be painted with polyurethane paint, Forest Green, Color Number 24052 of FED-STD-595. The paint shall be in accordance with MIL-C-83286, or shall be a commercial polyurethane paint which meets the following requirements of MIL-C-83286:

(1) Paragraph 3.7.3.4 Low temperature resistance.
(2) Paragraph 3.7.3.6.1 Accelerated weathering.
(3) Paragraph 3.7.3.7 Resistance to taping.

In addition, the paint shall have a minimum pencil hardness of 6H when tested in accordance with paragraph 4.8.12 of MIL-C-83286. All exterior trim items (such as wheels, wheel covers, running boards, metal radiator grilles, drip rails, bumpers, door handles, and light trim rings) shall be painted with the same paint as the exterior of the vehicle.

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3.4.2.2 Desert Sand. When specified (see 6.2), the outer surface of the vehicle and the interior surfaces of compartments (except in the cab) shall be painted with polyurethane paint in accordance with MIL-C-46168, Desert Sand, Color Number 30313 of FED-STD-595. All exterior trim items (such as wheels, wheel covers, running boards, metal radiator grilles, drip rails, bumpers, door handles, and light trim rings) shall be painted with the same paint as the exterior of the vehicle.

3.4.3 Rustproofing. Unless otherwise specified (see 6.2), the vehicle shall be rustproofed in accordance with the contractor's standard commercial practice. The first article units shall not be rustproofed until after approval of the first article test report.

3.5 Markings. Unless otherwise specified, markings for vehicles painted Forest Green shall be applied with black paint, Color Number 37038 of FED-STD-595, and markings for vehicles painted Desert Sand shall be applied with white paint, Color Number 37875 of FED-STD-595.

3.5.1 Identification marking. Unless otherwise specified (see 6.2), the vehicle shall be marked for the appropriate service in accordance with MIL-STD-1223.

3.6 Identification plates and operating instructions. All identification plates, instruction plates, data plates, etc, shall conform to MIL-P-514 and shall be of a size and shape consistent with the information required thereon. Abbreviations shall be in accordance with MIL-STD-12.

3.6.1 Identification plate. An identification plate, permanently marked in accordance with MIL-STD-130, shall be securely attached to the vehicle in a readily accessible location. The identification shall contain the following information:

NOMENCLATURE
MANUFACTURER'S MODEL NUMBER
MANUFACTURER'S SERIAL NUMBER
MANUFACTURER'S NAME
MANUFACTURER'S ADDRESS
MANUFACTURER'S SERVICE PHONE NUMBER
DATE OF MANUFACTURE
CONTRACT NUMBER
NATIONAL STOCK NUMBER

3.6.2 Control markings. All controls, valves, gauges, and indicators used in the operation of the vehicle shall be identified by nameplates securely installed in clearly visible locations.

3.6.3 Operating instructions. Brief operating and precautionary instructions shall be permanently affixed near the appropriate controls. The instructions shall be clear, concise, and adequate to enable operation of the vehicle without damage to the equipment or injury to personnel and shall refer to the components as identified by nameplates.

3.6.4 Transportation date plate. A transportation data plate shall be provided. The plate shall contain at least the following information:

a. Side and rear silhouette views of the vehicle.
b. Horizontal and vertical location of the center of gravity of the vehicle in transport condition.
c. Shipping weight.
d. Loading cubage.
e. Overall length, width, and height.
f. Tiedown instructions for air transport in Types C-130, C-141, C-17, and C-5 Aircraft.

3.7 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be marked for identification in accordance with MIL-STD-130. (Note that MIL-STD-130 does not require marking of commonly known

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commercial parts which present no identification problem or parts in assemblies which are not normally subject to disassembly or repair.

3.8 Dimensions. The vehicle, in fully operational condition, including all equipment and accessories, shall not exceed the following dimensions:

a. Length - 470 inches.
b. Width - 104 inches
c. Height - 101 inches.

3.9 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.4 and 6.2.1).

3.10 Fabrication.

3.10.1 Welders. All welders employed in the fabrication of the vehicle, except for commercial components, shall be certified in accordance with the applicable requirements of MIL-STD-248, the American Welding Society, or the American Society of Mechanical Engineers.

3.10.2 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface parts to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof, and service loadings. Ferrous and aluminum welding shall be in accordance with MIL-STD-248.

3.10.3 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Unless torque is specified, threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.10.4 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets shall be driven with pressure tools and shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape. Rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the component.

3.10.5 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots, or other irregularities. Where required for accurate adjustment, gear assemblies shall be free of backlash.

3.10.6 Cleaning. The vehicle shall be cleaned. Loose, spattered, or excess solder, welding slag, stray bolts, nuts and washers, rust, metal particles, pipe compound, and other foreign matter shall be removed during and after final assembly.

3.10.7 Soldering. Soldering shall be in accordance with MIL-STD-454, Requirement 5.

3.10.8 Brazing. Brazing shall be performed in accordance with MIL-B-7883.

3.11 Safety. All reciprocating, rotating, or moving parts of equipment shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces, including exhaust pipes and other lines which may be subject to high operating temperatures, exposed to contact by personnel or which create a fire hazard shall be fully guarded or insulated. Exhaust or discharges from equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard. Fuel tanks shall be located in a manner which will not allow spills and overflows to run onto the engine, exhaust, burners, or electrical equipment. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided. Steel walkways and ladders shall be provided where necessary for inspection, servicing, and maintenance. All platforms, steps, and ladder rungs shall have anti-skid surfaces. The truck-mounted mixer shall meet the applicable requirements of the Occupational Safety and Health Administration standards. A type B back-up alarm conforming to the performance criteria of SAE J994 shall be provided.

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3.12 Sound level. Sound level determinations shall be made in accordance with MIL-STD-1474. Noise levels in the cab shall not exceed 84 Db(A) with the door shut. If the noise level exceeds 84 Db(A) with the door wide open, a warning plate shall be permanently mounted in a conspicuous location. The label shall instruct the operator to wear hearing protection while operating the machine.

3.13 Workmanship. The vehicle, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp corners and edges; accuracy of dimensions, surface finish, and radii of fillets; thoroughness of soldering, welding, brazing, painting, wiring, and riveting; marking of parts and assemblies; alignment of parts and tightness of assembly fasteners, etc. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may use any inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that the vehicle conforms to the prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of Sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the vehicle shall be classified as follows:

a. First article testing (see 4.4 and 6.2.1).
b. Quality conformance inspection (see 4.5)

4.3 Test conditions.

4.3.1 Apparatus. Apparatus used in conjunction with the testing specified herein shall be of laboratory precision type and shall be calibrated at proper intervals to ensure laboratory accuracy.

4.3.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data and/or shorter intervals shall be provided as appropriate for any specific test.

a. Date.
b. Time started.
c. Time finished.
d. Ambient temperature.
e. Engine speed.
f. Engine oil pressure.

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g. Engine coolant temperature.
h. Transmission oil temperature.
i. Vehicle odometer reading at the start and finish of each test.
j. Engine hourmeter reading at the start and finish of each test.
k. Hydraulic fluid pressure at the pump outlet.
l. Hydraulic fluid temperature in the reservoir.

4.3.3 Preparation for tests. The vehicle, equipped with all specified auxiliary equipment, shall be fully serviced and all equipment shall be prepared for immediate operation.

4.3.4 Test rejection criteria. Throughout all tests specified herein, the vehicle shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein.
b. Spillage or leakage of any liquid, including fuel, coolant, lubricant, or hydraulic fluid, under any operating condition, except as allowed herein.
c. Structural failure of any component, including permanent deformation, or evidence of impending failure.
d. Evidence of excessive wear.
e. Interference between vehicle components or between the vehicle, the ground, and all required obstacles, with the exception of normal contact by the tires.
f. Misalignment of components.
g. Evidence of undesirable roadability characteristics, including instability in handling during cornering, braking, and while traversing rough terrain.
h. Conditions which present a safety hazard to personnel during operation, service, or maintenance.
i. Overheating of the engine, transmission, or any other vehicle component.

4.4 First article testing.

4.4.1 First article test sample. The first article test sample shall be identical to the items procured under the terms of the contract. Approval of the first article test report shall not constitute a waiver by the Government of its rights under the provisions of the contract.

4.4.2 First article test. The first article tests shall consist of all tests listed under 4.6.

4.4.3 Test report. The following shall be incorporated as a part of the test report:

a. All data collected, failures, maintenance performed, and other events recorded shall be identified by accumulated operation time, miles, cycles, or position in the test procedure, as appropriate. Test conditions at the time of failures or irregular operations identified shall be recorded.
b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.
c. Summary of the engineering reasoning behind any corrections made, to be made, or proposed to be made and the predicted effectiveness of those corrections.
d. List of all cleaning and finishing procedures to be used in accordance with MIL-STD-808.
e. List of special tools and diagnostic equipment necessary to perform field level maintenance.

4.4.4 Mobility tests.

4.4.4.1 Maximum speed and acceleration. The time-speed-distance relationship of the fully loaded vehicle shall be determined while accelerating from a standing start to maximum speed on a dry level paved surface.

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Not less than three runs, each in opposite directions, shall be conducted. Maximum speed and acceleration shall be the average for all runs.

4.4.4.2 Grade operation. The fully loaded vehicle shall be driven up and down a 25 percent dry paved grade. The maximum speed that can be maintained while climbing the grade shall be measured and recorded in the test report. Engine oil pressure shall be observed and shall not be less than the manufacturer's recommended minimum pressure at the engine operating speed. Engine starting ability shall be demonstrated in both directions on the grade.

4.4.4.3 Side slope operation. The fully loaded vehicle shall be driven in both directions on a side slope of 20 percent. Engine oil pressure shall be observed and shall not be less than the manufacturer's recommended minimum pressure at the engine operating speed. Engine starting ability shall be demonstrated in both directions on the slope.

4.4.4.4 Ramp test. The fully loaded vehicle shall be driven forward from a hard flat horizontal surface and ascend a ramp with a minimum length of 1.2 times the vehicle wheelbase and a slope of 17°. It shall then proceed across the ramp crest onto an upper connecting hard horizontal surface until all tires are resting on the upper surface, down the ramp, and onto the lower horizontal surface. The test shall then be repeated in the opposite direction.

4.4.4.5 Brake tests.

4.4.4.5.1 Service brake tests. The vehicle shall be tested in accordance with SAE J786a.

4.4.4.5.2 Parking brake tests. The vehicle shall be tested in accordance with SAE J360.

4.4.4.6 Wall to wall turning diameter. The vehicle wall to wall turning diameter for turns in each direction shall be measured in accordance with SAE J695.

4.4.5 Capacity testing.

4.4.5.1 Loading. Demonstrate the ability of the mixer bins to be loaded by a front end loader by filling the sand and aggregate bins with sand and aggregate respectively with a front end loader. Inability of the mixer bins to be loaded by the front end loader shall constitute failure of this test.

4.4.5.2 Mounting. With all bins and the water tank loaded to capacity, test the mounting of the mixer on the chassis by accelerating from stop to 10 miles per hour as quickly as possible, then applying brakes to stop suddenly. Repeat this 10 times. Accelerate backward from rest to maximum reverse speed as quickly as possible, then apply brakes to stop suddenly. Repeat this 10 times. Evidence of permanently deformed or broken mounting devices or misalignment between chassis and mixer shall constitute failure of this test. Mixer shall be transported over paved roads at speeds up to 55 mph for a distance of 30 miles while fully loaded. Mixer shall be transported over unimproved dirt or gravel for a period of fifteen minutes at a maximum safe speed. Permanent deformation, overheating, malfunction, failure to complete and test, or leakage of lubricants or hydraulic fluids shall be cause for rejection of the first article.

4.4.5.3 Mixing ability. Demonstrate the mixing ability of the mixer by setting the controls for a six bag mix and producing a minimum of one cubic yard of concrete. Examine the fresh concrete to determine if it is a uniform mix. Examine the aggregate for complete coating with cement paste. Inability to produce a uniform mix or to completely coat the aggregate shall constitute failure of this test. Inability to produce one cubic yard of five bag mix concrete within four minutes shall constitute failure of this test. Demonstrate the ability of the mixer to produce various design strength mixes by setting the controls for a four bag mix and producing a minimum of one cubic yard of concrete. Examine the fresh concrete to determine if it is a uniform mix. Examine the aggregate for complete coating with cement paste. Inability of the mixer to produce a uniform mix or to completely coat the aggregate shall constitute failure of this test.

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4.4.5.4 Axle loading. Determine, by weighing, the load on each individual axle under the following conditions:

a. Water supply tank filled to capacity.
b. Coarse aggregate bin filled to capacity. (weight of the aggregate shall be not less than 100 pounds/cu. ft.)
c. Sand bin filled to capacity. (weight of sand shall be not less than 95 pounds/cubic foot)
d. Cement bin filled to capacity.
e. Wet and dry admixture systems filled to capacity.
f. All reservoirs, to include fuel tank(s), filled to capacity.

The weight on each axle shall not exceed the recommended weight for the truck chassis class. If the axles are overloaded, this constitutes failure of this test.

4.4.5.5 Test procedures for slinging provisions. Test procedures for slinging provisions shall be as follows:

a. Lift the truck-mounted mixer and hold in suspension in its normal travel position using slings that cover not more than 24 feet above the lowest extremity of the truck-mounted mixer. Determine the true angle of the sling legs from the vertical. Measure the clearance between each sling and the truck-mounted mixer. Measure eye openings and clearance dimensions of the provisions. Measure the static load carried by each provision.
b. Restrain the truck-mounted mixer by anchoring the main frame and subject each provision to a working load force equal to the load factor specified in MIL-STD-209 times the load it carried in "a" above. An alternate method of test is to lift the truck-mounted mixer as in "a" above and add weights to the main frame until the required force is obtained. Hold the truck-mounted mixer under either of the above loading conditions for at least 90 seconds.

4.4.5.6 Test procedures of equipment tiedown provisions. With the truck-mounted mixer anchored by means other than the provisions being tested, apply the distributed loads as specified in MIL-STD-209 to each attachment. Measure dimensions of the tiedown.

4.4.6 Design verification.

4.4.6.1 Dimensional verification. The dimensions specified herein shall be measured with the vehicle standing on a flat level surface. Overall height and vehicle lean shall be measured in both the empty and the fully loaded conditions.

4.4.6.2 Sound level testing. The vehicle shall be tested in accordance with SAE J336a, J366b, and J1077 to demonstrate compliance with 3.13.

4.4.7 Environmental tests.

4.4.7.1 Low temperature storage and operation. The complete vehicle, including all accessories, shall be tested in accordance with Test Method 502.2, Procedures I and II, of MIL-STD-810.. Temperature sensor locations shall be approved by the procuring activity. Operational checkouts shall be performed in accordance with paragraph 4.4.7.4. When applicable, a fine mist of water shall be applied to the entire windshield to form a thin film of ice. The defroster shall clear the windshield within one hour and maintain clear vision throughout the remainder of the test. "Standard ambient" and "controlled ambient" shall be defined as the ambient condition of the test site. For Procedure I, the vehicle shall be exposed to the minimum required cold storage temperature of -25°F in a steady state condition for 24 hours. For Procedure II, the vehicle shall be exposed to the minimum required cold operational temperature of -25°F in a steady state condition.

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4.4.7.2 High temperature storage and operation. The complete vehicle, including all accessories, shall be tested in accordance with Test Method 501.2, Procedures I and II, of MIL-STD-810 to demonstrate compliance with 3.1.5. Temperature sensor locations shall be approved by the procuring activity. Operational checkouts shall be performed in accordance with 4.4.7.4. "Standard ambient" and "controlled ambient" shall be defined as the ambient condition of the test site. For Procedure I, the vehicle shall be exposed to the maximum required hot storage temperature of +125°F in a steady state condition for 24 hours. For Procedure II, the vehicle shall be exposed to the maximum required hot operational temperature of +125°F in a steady state condition.

4.4.7.3 Rain. The complete vehicle, including all accessories, shall be tested in accordance with Test Method 506.2, Procedure III, of MIL-STD-810. Operational checkouts shall be in accordance with 4.4.7.4. The interior of the cab and all compartments shall be examined for water; any evidence of leakage shall be cause for rejection.

4.4.7.4 Operational checkout. All controls, levers, mechanisms, and doors shall be actuated a sufficient number of times to ensure consistency of operation and to determine whether the test exposure produced system degradation.

4.4.8 Maintainability demonstration. Compliance with the repair time requirements shall be demonstrated by task performance. All scheduled maintenance tasks and all tasks listed in 3.1.4.3 shall be performed. In addition, corrective maintenance tasks shall be selected by the procuring activity from an approved list of tasks provided by the contractor in the first article test procedure. This list shall represent all expected failures over the life of the vehicle. Not less than 25 or more than 50 of the tasks shall be demonstrated to verify a maximum repair time within the required limit. As part of this demonstration, the recommended frequencies of the scheduled maintenance tasks and the times recorded to accomplish the tasks shall be used to develop an expected value of scheduled maintenance time per measure of use, such as calendar time, hours of operation, or miles traveled. The capability of performing maintenance tasks by personnel wearing arctic mittens shall be demonstrated.

4.4.9 Air transportability analysis. A detailed engineering analysis shall be conducted to demonstrate compliance with 3.1.6. The analysis shall include the tiedowns and all major components and their ability to withstand the accelerations specified in MIL-STD-1791. The evaluation shall also include a dimensional analysis for the vehicle while loaded aboard Types C-130, C-141, C-17, and C-5 Aircraft.

4.5 Quality conformance inspection.

4.5.1 Examination. Each truck-mounted mixer shall be examined for the following characteristics. Presence of one or more defects shall be cause for rejection.

101. Materials not as specified.	3.1.3
102. Design and construction not as specified.	3.1.4
103. Reliability not as specified.	3.1.4.1
104. Maintainability not as specified.	3.1.4.2
105. Ease of maintenance not as specified.	3.1.4.3
106. Fastening devices not as specified.	3.1.4.4
107. Lubrication not as specified.	3.1.4.6
108. Hydraulic fluid not as specified.	3.1.4.7
109. Air transportability not as specified.	3.1.6
110. Tiedown attachments not as specified.	3.1.6.2
111. Sling attachments not as specified.	3.1.6.3
112. Mixer construction not as specified.	3.2
113. Bins and tanks not as specified.	3.2.3
114. Hopper not as specified.	3.2.4
115. Water system not as specified.	3.2.5

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116. Mixing system not as specified.	3.2.6
117. Mix proportioning system not as specified.	3.2.7
118. Mixer drive train not as specified.	3.2.8
119. Quick loading cement bin not as specified.	3.2.9
120. Admixture system not as specified.	3.2.10
121. Air system not as specified.	3.2.11
122. Truck chassis not as specified (KKK-T-2109).	3.3
123. Treatment and painting not as specified.	3.4
124. Markings not as specified.	3.5
125. Identification and instruction plate not as specified.	3.6
126. Interchangeability not as specified.	3.7
127. Dimensions not as specified.	3.8
128. Welding not as specified.	3.10.2
129. Bolted connections not as specified.	3.10.3
130. Riveted connections not as specified.	3.10.4
131. Gear and lever assembly not as specified.	3.10.5
132. Soldering not as specified.	3.10.7
133. Brazing not as specified.	3.10.8
134. Workmanship not as specified.	3.11
135. Safety not as specified.	3.12
136. Sound level not as specified.	3.13

4.6 Test methods.

4.6.1 Examination of product. The vehicle shall be inspected to determine compliance with the requirements specified herein. Particular attention shall be given to materials, workmanship, dimensions, surface finishes, protective coating and their application, welding and riveting, and markings. Any painted surface on the exterior of the vehicle may be tested in accordance with paragraph 4.8.10 of MIL-C-83286.

4.6.2 Inspection comparison. The Government may select truck-mounted mixers at any time during the contract production period and subject these truck-mounted mixers to the examination and the tests specified in this purchase description to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction truck-mounted mixer and initial production truck-mounted mixer. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.3 Inspection failure. Failure of an inspection comparison truck-mounted mixer to meet any requirement specified herein during and as a result of the examination and tests specified in this purchase description shall be cause for refusal by the Government to continue acceptance of production until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all truck-mounted mixers accepted prior to completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Refurbishment. After successful completion of the all required testing, each vehicle shall be restored to like new condition.

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5. PACKAGING.

5.1 Preparation for delivery. Preparation for delivery shall be in accordance with the terms of the procuring activity.

6. NOTES.
(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The truck-mounted mixer is intended for use in mixing concrete at a remote area.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

Finish color required, if not Forest Green, Color Number 24052 of FED-STD-595 (see 3.4.2).

Rustproofing required, if other than the manufacturer's standard commercial practice (see 3.4.3).

Identification markings required (see 3.5.1).

b. KKK-T-2109 Ordering data applies to the truck chassis on which the mixer is mounted and must be obtained from the contract, or from the contracting officer upon request.

6.2.1 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of the first article. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.3 Data Requirements. The following Data Item Descriptions (DIDs) must be listed, as applicable, on the Contract Data Requirements List (DD Form 1423) when this specification is applied on a contract, in order to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

DID Number	DID Title	Suggested Tailoring
DI-T-4901	FAT Procedures	See DD Form 1423
DI-T-4902	FA Inspection Report	See DD Form 1423
DI-T-2373A/T	Notification of Test	See DD Form 1423

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Forms 1423.

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6.4 Subject term (key word)listing.

Aggregate
Bins
Hopper
Mean-time-between-failure (MTBF)
Pneumatic
Tank

Custodians:
Air Force - 99

Preparing activity:
Air Force - 84

Project No: 3895-F267

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NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-M-87996(84)	Mixer, Concrete, Mobile, 5 Cubic Yard Truck Mounted

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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